

```
import torch

print(torch.cuda.is_available())
print(torch.cuda.get_device_name(0))
```

```
True
Tesla T4
```

```
from google.colab import files

uploaded = files.upload()
```

[Choose Files](#) Project5\_A...detection.zip

**Project5\_Ag\_Crop and weed detection.zip**(application/x-zip-compressed) - 83634945 bytes, last modified: 6/14/2026 - 100% done  
Saving Project5\_Ag\_Crop and weed detection.zip to Project5\_Ag\_Crop and weed detection (1).zip

```
import zipfile
import os

zip_path = "Project5_Ag_Crop and weed detection.zip"

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall("/content/dataset")

print("Dataset Extracted")
```

```
Dataset Extracted
```

```
import os

for root, dirs, files in os.walk("/content/dataset"):
    print(root)
```

```
/content/dataset
/content/dataset/Project5_Ag_Crop and weed detection
```

```
!pip install ultralytics -q
```

```

_____ 41.3/41.3 kB 2.5 MB/s eta 0:00:00
_____ 1.3/1.3 MB 35.2 MB/s eta 0:00:00
```

```
import os

for root, dirs, files in os.walk("/content"):
    if len(files) > 0:
        print(root)
```

```
/content
/content/.config
/content/.config/logs/2026.06.04
/content/.config/configurations
/content/dataset/Project5_Ag_Crop and weed detection
/content/sample_data
```

```
!ls "/content/dataset/Project5_Ag_Crop and weed detection"
```

```
'~$object1.docx'
'Project5_Ag_Crop and weed detection.docx'
'Project5_Ag_Crop and weed detection.zip'
```

```
import zipfile

inner_zip = "/content/dataset/Project5_Ag_Crop and weed detection/Project5_Ag_Crop and weed detection.zip"

with zipfile.ZipFile(inner_zip, 'r') as zip_ref:
    zip_ref.extractall("/content/agri_dataset")

print("Inner ZIP extracted successfully!")
```

```
Inner ZIP extracted successfully!
```

```
!find /content/agri_dataset -type d
```

```
/content/agri_dataset
/content/agri_dataset/agri_data
/content/agri_dataset/agri_data/data
```

```
import os
```

```

import random
import shutil

SOURCE = "/content/agri_dataset/agri_data/data"

# Create folders
os.makedirs("/content/agri/images/train", exist_ok=True)
os.makedirs("/content/agri/images/val", exist_ok=True)

os.makedirs("/content/agri/labels/train", exist_ok=True)
os.makedirs("/content/agri/labels/val", exist_ok=True)

# Get image files
images = [f for f in os.listdir(SOURCE)
           if f.lower().endswith((".jpg", ".jpeg", ".png"))]

print("Total Images Found:", len(images))

# Shuffle dataset
random.shuffle(images)

# 80-20 split
split_index = int(len(images) * 0.8)

train_images = images[:split_index]
val_images = images[split_index:]

# Copy training files
for img in train_images:

    label = os.path.splitext(img)[0] + ".txt"

    shutil.copy(
        os.path.join(SOURCE, img),
        "/content/agri/images/train"
    )

    if os.path.exists(os.path.join(SOURCE, label)):
        shutil.copy(
            os.path.join(SOURCE, label),
            "/content/agri/labels/train"
        )

# Copy validation files
for img in val_images:

    label = os.path.splitext(img)[0] + ".txt"

    shutil.copy(
        os.path.join(SOURCE, img),
        "/content/agri/images/val"
    )

    if os.path.exists(os.path.join(SOURCE, label)):
        shutil.copy(
            os.path.join(SOURCE, label),
            "/content/agri/labels/val"
        )

print("=====")
print("Training Images :", len(train_images))
print("Validation Images :", len(val_images))
print("Dataset Split Complete")

```

```

Total Images Found: 1300
=====
Training Images : 1040
Validation Images : 260
Dataset Split Complete

```

```

print("Train Images:",
      len(os.listdir("/content/agri/images/train")))

print("Val Images:",
      len(os.listdir("/content/agri/images/val")))

print("Train Labels:",
      len(os.listdir("/content/agri/labels/train")))

print("Val Labels:",
      len(os.listdir("/content/agri/labels/val")))

```

```
Train Images: 1040
Val Images: 260
Train Labels: 1040
Val Labels: 260
```

```
yaml_text = """
path: /content/agri

train: images/train
val: images/val

nc: 2

names:
  0: crop
  1: weed
"""

with open("/content/agri/data.yaml", "w") as f:
    f.write(yaml_text)

print("data.yaml created successfully")
```

```
data.yaml created successfully
```

```
with open("/content/agri/data.yaml") as f:
    print(f.read())
```

```
path: /content/agri

train: images/train
val: images/val

nc: 2

names:
  0: crop
  1: weed
```

```
!pip install ultralytics -q
```

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

model.train(
    data="/content/agri/data.yaml",
    epochs=50,
    imgsz=512,
    batch=16
)
```

```

0.88188, 0.88288, 0.88388, 0.88488, 0.88589, 0.88689, 0.88789,
0.88889, 0.88989, 0.89089, 0.89189, 0.89289, 0.89389, 0.89489, 0.8959,
0.8969, 0.8979, 0.8989, 0.8999, 0.9009, 0.9019, 0.9029, 0.9039, 0.9049,
0.90591, 0.90691, 0.90791, 0.90891, 0.90991, 0.91091, 0.91191,
0.91291, 0.91391, 0.91491, 0.91592, 0.91692, 0.91792, 0.91892, 0.91992,
0.92092, 0.92192, 0.92292, 0.92392, 0.92492, 0.92593, 0.92693, 0.92793, 0.92893,
0.92993, 0.93093, 0.93193, 0.93293, 0.93393, 0.93493, 0.93594,
0.93694, 0.93794, 0.93894, 0.93994, 0.94094, 0.94194, 0.94294, 0.94394,
0.94494, 0.94595, 0.94695, 0.94795, 0.94895, 0.94995, 0.95095, 0.95195, 0.95295,
0.95395, 0.95495, 0.95596, 0.95696, 0.95796, 0.95896, 0.95996,
0.96096, 0.96196, 0.96296, 0.96396, 0.96496, 0.96597, 0.96697, 0.96797,
0.96897, 0.96997, 0.97097, 0.97197, 0.97297, 0.97397, 0.97497, 0.97598, 0.97698,
0.97798, 0.97898, 0.97998, 0.98098, 0.98198, 0.98298, 0.98398,
0.98498, 0.98599, 0.98699, 0.98799, 0.98899, 0.98999, 0.99099, 0.99199,
0.99299, 0.99399, 0.99499, 0.996, 0.997, 0.998, 0.999, 1]], array([[
0.99598, 0.99598, 0.99598, ..., 0, 0, 0],
[ 0.97423, 0.97423, 0.96392, ..., 0, 0, 0]]), 'Confidence', 'Recall'])
fitness: 0.5661444106338634
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)', 'metrics/mAP50-95(B)']
maps: array([ 0.57415, 0.55814])
names: {0: 'crop', 1: 'weed'}
nt_per_class: array([249, 194])
nt_per_image: array([113, 148])
results_dict: {'metrics/precision(B)': 0.8156704739452523, 'metrics/recall(B)': 0.7832877903366041, 'metrics/mAP50(B)':
0.8497987490669047, 'metrics/mAP50-95(B)': 0.5661444106338634, 'fitness': 0.5661444106338634}
save_dir: PosixPath('/content/runs/detect/train')
speed: {'preprocess': 0.19542458076802388, 'inference': 2.1844483576912284, 'loss': 0.000526269230669221, 'postprocess':

```

```

import os

print("Train Images:",
      len(os.listdir("/content/agri/images/train")))

print("Val Images:",
      len(os.listdir("/content/agri/images/val")))

print("Train Labels:",
      len(os.listdir("/content/agri/labels/train")))

print("Val Labels:",
      len(os.listdir("/content/agri/labels/val")))

```

```

Train Images: 1040
Val Images: 260
Train Labels: 1040
Val Labels: 260

```

```

from ultralytics import YOLO

model = YOLO("/content/runs/detect/train/weights/best.pt")

metrics = model.val()

print(metrics)

```

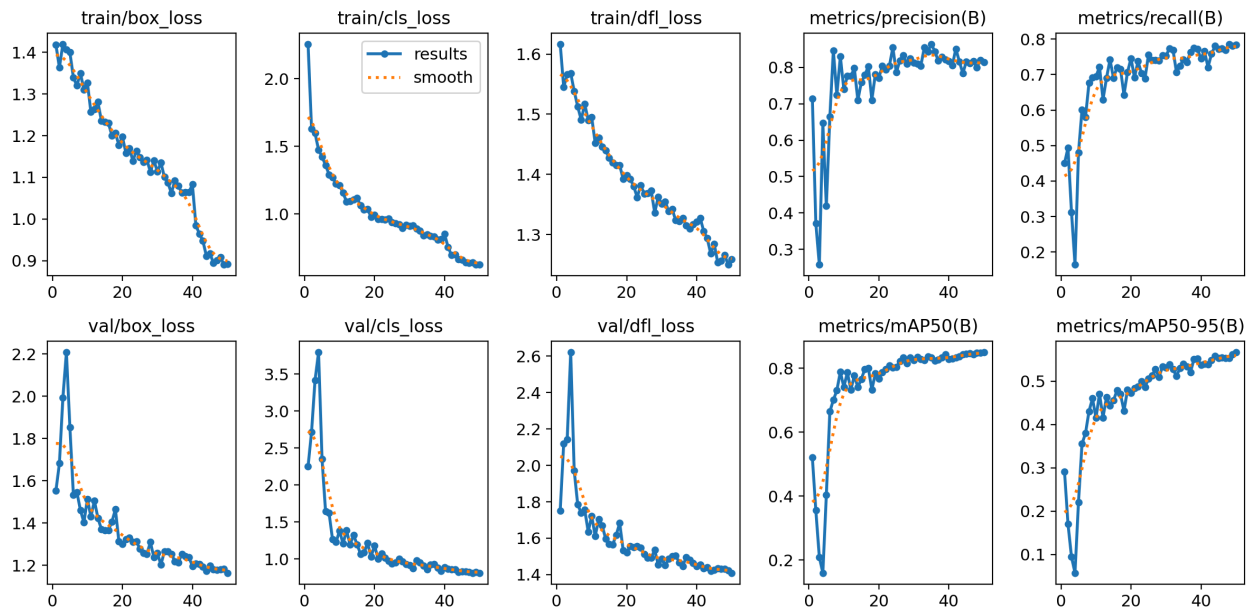
```

0.64865, 0.64965, 0.65065, 0.65165, 0.65265, 0.65365, 0.65465, 0.65566, 0.65666
0.67267, 0.67367, 0.67467, 0.67568, 0.67668, 0.67768, 0.67868, 0.67968, 0.68068
0.6967, 0.6977, 0.6987, 0.6997, 0.7007, 0.7017, 0.7027, 0.7037, 0.7047
0.72072, 0.72172, 0.72272, 0.72372, 0.72472, 0.72573, 0.72673, 0.72773, 0.72873
0.74474, 0.74575, 0.74675, 0.74775, 0.74875, 0.74975, 0.75075, 0.75175, 0.75275
0.76877, 0.76977, 0.77077, 0.77177, 0.77277, 0.77377, 0.77477, 0.77578, 0.77678
0.79279, 0.79379, 0.79479, 0.7958, 0.7968, 0.7978, 0.7988, 0.7998, 0.8008
0.81682, 0.81782, 0.81882, 0.81982, 0.82082, 0.82182, 0.82282, 0.82382, 0.82482
0.84084, 0.84184, 0.84284, 0.84384, 0.84484, 0.84585, 0.84685, 0.84785, 0.84885
0.86486, 0.86587, 0.86687, 0.86787, 0.86887, 0.86987, 0.87087, 0.87187, 0.87287
0.88889, 0.88989, 0.89089, 0.89189, 0.89289, 0.89389, 0.89489, 0.8959, 0.8969
0.91291, 0.91391, 0.91491, 0.91592, 0.91692, 0.91792, 0.91892, 0.91992, 0.92092
0.93694, 0.93794, 0.93894, 0.93994, 0.94094, 0.94194, 0.94294, 0.94394, 0.94494
0.96096, 0.96196, 0.96296, 0.96396, 0.96496, 0.96597, 0.96697, 0.96797, 0.96897
0.98498, 0.98599, 0.98699, 0.98799, 0.98899, 0.98999, 0.99099, 0.99199, 0.99299
[ 0.97423, 0.97423, 0.96392, ..., 0, 0, 0]]], 'Confidence', 'Recall']]
fitness: 0.5670035055049629
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)', 'metrics/mAP50-95(B)']
maps: array([ 0.57547, 0.55854])
names: {0: 'crop', 1: 'weed'}
nt_per_class: array([249, 194])
nt_per_image: array([113, 148])
results_dict: {'metrics/precision(B)': 0.8148684118842311, 'metrics/recall(B)': 0.7832877903366041, 'metrics/mAP50(B)': 0.
save_dir: PosixPath('/content/runs/detect/val')
speed: {'preprocess': 1.7426846500012079, 'inference': 3.5851418384639215, 'loss': 0.0007321269234895365, 'postprocess': 1
state: {'tn': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img': []}

```

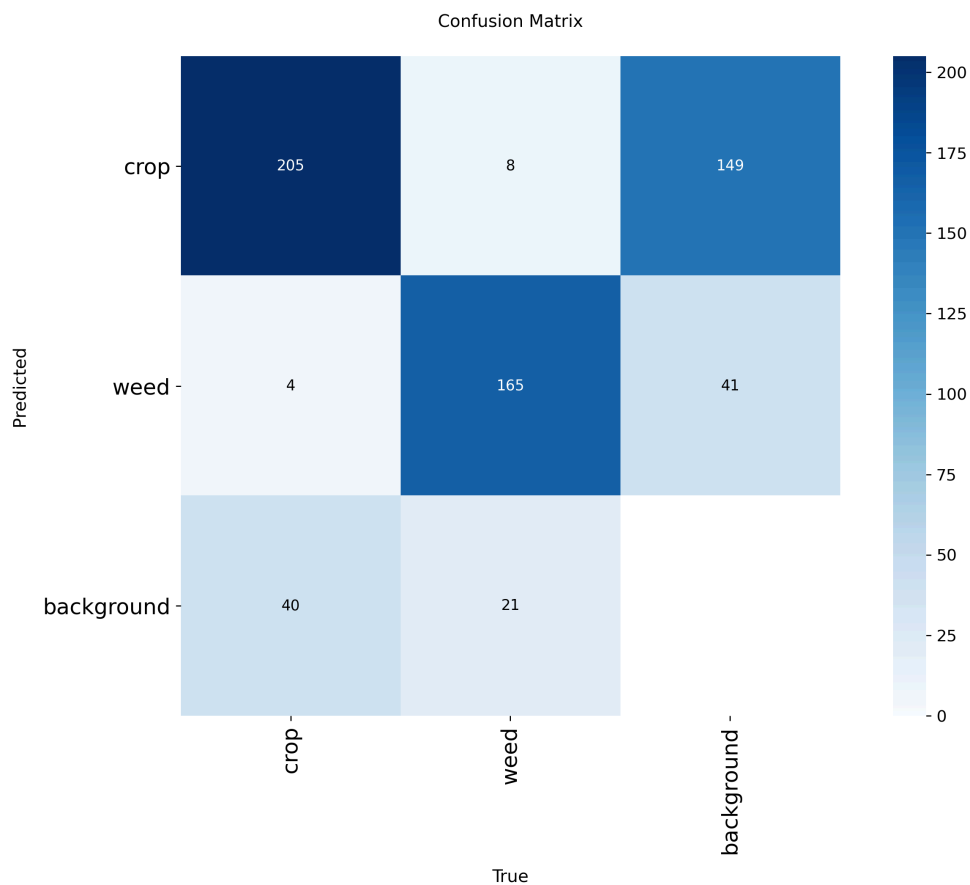
```
from IPython.display import Image, display
```

```
display(Image("/content/runs/detect/train/results.png"))
```



```
from IPython.display import Image, display
```

```
display(Image("/content/runs/detect/train/confusion_matrix.png"))
```



```
from google.colab import files
```

```
uploaded = files.upload()
```

agri\_0\_909.jpeg

**agri\_0\_909.jpeg**(image/jpeg) - 49502 bytes, last modified: 4/6/2020 - 100% done  
Saving agri\_0\_909.jpeg to agri\_0\_909.jpeg

```
from ultralytics import YOLO
```

```
model = YOLO("/content/runs/detect/train/weights/best.pt")
```

```
results = model.predict(
    source="agri_0_909.jpeg",
    conf=0.25,
    save=True
)
```

image 1/1 /content/agri\_0\_909.jpeg: 512x512 1 weed, 8.7ms  
Speed: 1.2ms preprocess, 8.7ms inference, 1.4ms postprocess per image at shape (1, 3, 512, 512)  
Results saved to /content/runs/detect/predict

```
import os
```

```
predict_folder = "/content/runs/detect/predict"
```

```
print(os.listdir(predict_folder))
```

```
['agri_0_909.jpg']
```

```
import os
```

```
for root, dirs, files in os.walk("/content/runs/detect"):
    print(root)
```

```
/content/runs/detect
/content/runs/detect/train
/content/runs/detect/train/weights
/content/runs/detect/val
/content/runs/detect/predict
```

```
!ls /content/runs/detect
```

```
predict train val
```

```
!ls /content/runs/detect/predict2
```

```
ls: cannot access '/content/runs/detect/predict2': No such file or directory
```

```
!ls /content/runs/detect/predict
```

```
agri_0_909.jpg
```

```
import os
from IPython.display import Image, display

predict_dirs = sorted(
    [d for d in os.listdir("/content/runs/detect")
     if d.startswith("predict")]
)

latest_predict = predict_dirs[-1]

folder = f"/content/runs/detect/{latest_predict}"

print("Using folder:", folder)

files = os.listdir(folder)

image_files = [f for f in files
                if f.lower().endswith((".jpg", ".jpeg", ".png"))]

print("Found:", image_files)

display(Image(os.path.join(folder, image_files[0])))
```

```
Using folder: /content/runs/detect/predict
Found: ['agri_0_909.jpg']
```



```
from ultralytics import YOLO

model = YOLO("/content/runs/detect/train/weights/best.pt")

results = model("agri_0_909.jpeg")
```

```

crop_count = 0
weed_count = 0

for box in results[0].boxes:
    cls = int(box.cls)

    if cls == 0:
        crop_count += 1
    elif cls == 1:
        weed_count += 1

print("Crop Count:", crop_count)
print("Weed Count:", weed_count)

```

```

image 1/1 /content/agri_0_909.jpeg: 512x512 1 weed, 7.5ms
Speed: 1.6ms preprocess, 7.5ms inference, 1.2ms postprocess per image at shape (1, 3, 512, 512)
Crop Count: 0
Weed Count: 1

```

```
!find /content/runs/detect -name "agri_0_909*"
```

```
/content/runs/detect/predict/agri_0_909.jpg
```

```

from IPython.display import Image, display

display(Image("/content/runs/detect/predict/agri_0_909.jpg"))

```



```

from google.colab import files

files.download("/content/runs/detect/train/weights/best.pt")

```

```

from google.colab import files

files.download("/content/runs/detect/train/results.png")
files.download("/content/runs/detect/train/confusion_matrix.png")

```

```

from ultralytics import YOLO

model = YOLO("/content/runs/detect/train/weights/best.pt")

metrics = model.val()

```

```

Ultralytics 8.4.68 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 3,006,038 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (ping: 0.0±0.0 ms, read: 1510.5±411.1 MB/s, size: 64.5 KB)
val: Scanning /content/agri/labels/val.cache... 260 images, 0 backgrounds, 0 corrupt: 100% 260/260 45.4Mit/s 0.
Class  Images  Instances  Box(P  R  mAP50  mAP50-95): 100% 17/17 5.1it/s 3.
all      260      443      0.815  0.783  0.85  0.567

```



|      |     |     |       |       |       |       |
|------|-----|-----|-------|-------|-------|-------|
| crop | 113 | 249 | 0.719 | 0.747 | 0.826 | 0.575 |
| weed | 148 | 194 | 0.911 | 0.82  | 0.874 | 0.559 |

Speed: 1.7ms preprocess, 3.0ms inference, 0.0ms loss, 1.4ms postprocess per image  
Results saved to `/content/runs/detect/val-2`

```
from google.colab import files
```

```
files.download("/content/runs/detect/train/weights/best.pt")
```

```
files.download("/content/runs/detect/train/results.png")
```

```
files.download("/content/runs/detect/train/confusion_matrix.png")
```

```
files.download("/content/runs/detect/train/labels.jpg")
```

```
import os
from ultralytics import YOLO

model = YOLO("/content/runs/detect/train/weights/best.pt")

test_images = os.listdir("/content/agri/images/val")

for img in test_images[:5]:
    print(f"\nTesting: {img}")
    results = model(f"/content/agri/images/val/{img}")

    crop = 0
    weed = 0

    for box in results[0].boxes:
        cls = int(box.cls)

        if cls == 0:
            crop += 1
        elif cls == 1:
            weed += 1

    print("Crop:", crop)
    print("Weed:", weed)
```

Testing: agri\_0\_4560.jpeg

image 1/1 /content/agri/images/val/agri\_0\_4560.jpeg: 512x512 1 weed, 8.1ms  
Speed: 1.3ms preprocess, 8.1ms inference, 1.8ms postprocess per image at shape (1, 3, 512, 512)  
Crop: 0  
Weed: 1

Testing: agri\_0\_2034.jpeg

image 1/1 /content/agri/images/val/agri\_0\_2034.jpeg: 512x512 1 crop, 6.2ms  
Speed: 1.1ms preprocess, 6.2ms inference, 1.1ms postprocess per image at shape (1, 3, 512, 512)  
Crop: 1  
Weed: 0

Testing: agri\_0\_9047.jpeg

image 1/1 /content/agri/images/val/agri\_0\_9047.jpeg: 512x512 2 crops, 5.7ms  
Speed: 1.0ms preprocess, 5.7ms inference, 1.1ms postprocess per image at shape (1, 3, 512, 512)  
Crop: 2  
Weed: 0

Testing: agri\_0\_4502.jpeg

image 1/1 /content/agri/images/val/agri\_0\_4502.jpeg: 512x512 1 weed, 6.0ms  
Speed: 1.4ms preprocess, 6.0ms inference, 1.2ms postprocess per image at shape (1, 3, 512, 512)  
Crop: 0  
Weed: 1

Testing: agri\_0\_2533.jpeg

image 1/1 /content/agri/images/val/agri\_0\_2533.jpeg: 512x512 19 crops, 5.6ms  
Speed: 0.9ms preprocess, 5.6ms inference, 1.1ms postprocess per image at shape (1, 3, 512, 512)  
Crop: 19  
Weed: 0

